

AND Gate

Truth table

AND Gate		
a	b	y
0	0	0
0	1	0
1	0	0
1	1	1

Behavioral source code (and_gate.v)

```
module and_gate(y,a,b);
output y;
input a,b;
assign y=a&b;
endmodule
```

Testbench source code (and_gate_tb.v)

```
`timescale 1ns/1ps

module and_gate_tb;
    // 1. Creating registers for inputs and a wire for the output
    reg test_a;
    reg test_b;
    wire test_y;

    // 2. Instantiating the actual AND gate module (connect pins)
    and_gate uut (
        .y(test_y),
        .a(test_a),
        .b(test_b)
    );
```

```
// 3. Generating the input signals over time
initial begin
    // Telling iverilog to dump waveform data for GTKWave
    $dumpfile("simulation.vcd");
    $dumpvars(0, and_gate_tb);

    // Test case 1- a=0,b=0
    test_a=0; test_b=0; #10;
    // Test case 2- a=0,b=1
    test_a=0; test_b=1; #10;
    // Test case 3- a=1,b=0
    test_a=1; test_b=0; #10;
    // Test case 4- a=1,b=1
    test_a=1; test_b=1; #10;

    $finish; // End simulation
end
endmodule
```

fgfg

```
call C:\oss-cad-suite\environment.bat
```

fgfg

```
iverilog -o simulation.vvp and_gate.v and_gate_tb.v
```

fgfg

```
vvp simulation.vvp
```

Generate & check output waveform - Fig. 1(a)

```
gtkwave simulation.vcd
```

```
yosys
```

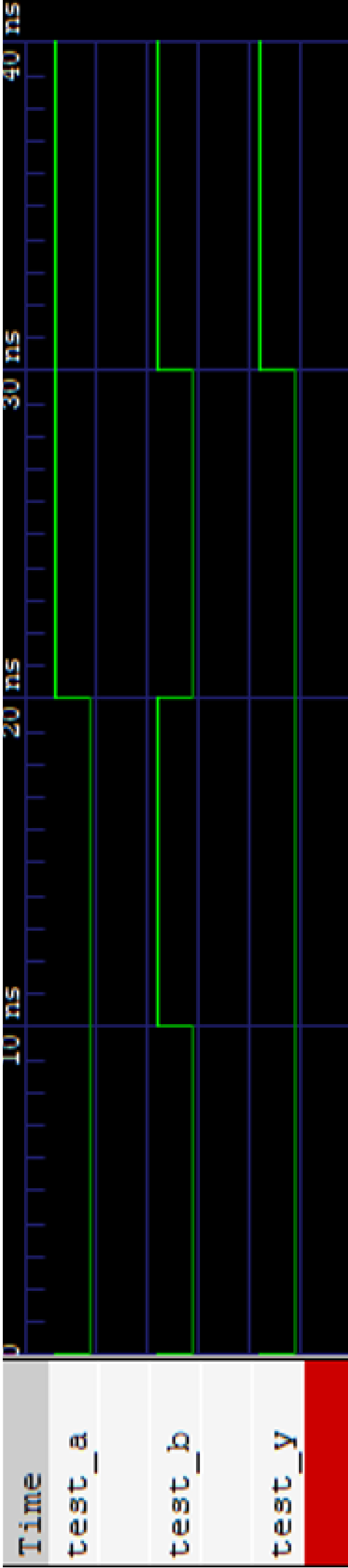


Fig. 2(a) - AND gate output waveform

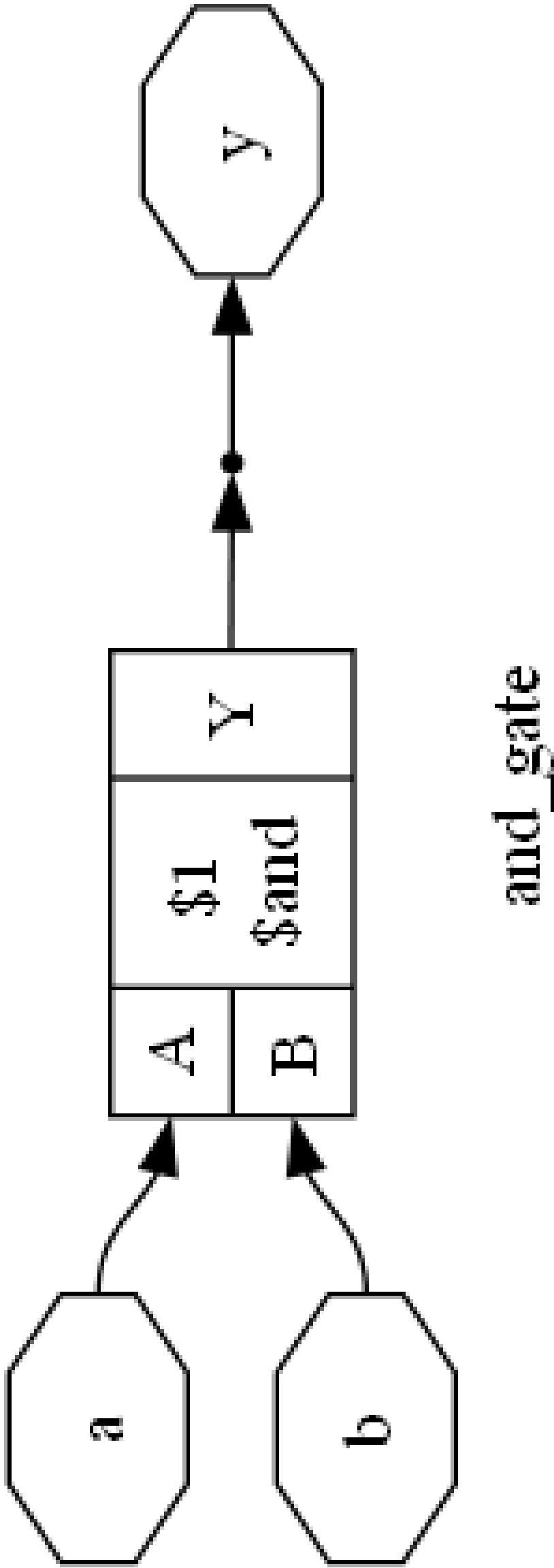


Fig. 2(b) - AND gate generated schematic

```
/-----\
| yosys -- Yosys Open SYnthesis Suite                      |
| Copyright (C) 2012 - 2026  Claire Xenia Wolf <claire@yosyshq.com> |
| Distributed under an ISC-like license, type "license" to see terms |
\-----/
Yosys 0.68+40 (git sha1 0f2bcb94b-dirty, Release, GNU /usr/bin/x86_64-w64-mingw32-g++ 15.2.1)
```

```
yosys> read_verilog and_gate.v
```

```
yosys> hierarchy -top and_gate
```

```
yosys> proc opt techmap opt
```

```
yosys> write_verilog and_gate_netlist.v
```

```
yosys> show
```

```
5. Generating Graphviz representation of design.
Writing dot description to 'show.dot'.
Dumping module device to page 1.
```

```
yosys> exit
```

```
Warnings: 2 unique messages, 3 total
End of script. Logfile hash: a730de9a18
Yosys 0.68+40 (git sha1 0f2bcb94b-dirty, Release, GNU /usr/bin/x86_64-w64-mingw32-g++ 15.2.1)
Time spent: 5% 2x write_verilog (0 sec), 5% 2x read_verilog (0 sec), ...
```

```
[OSS CAD Suite] D:\ XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX \AND>"D:\ XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX \windows_10_cmake_Release_Graphviz-15.1.1-win64\Graphviz-
15.1.1-win64\bin\dot.exe" -Tpng show.dot -o schematic.png
```

Generate & check and_gate schematic - Fig. 2(b)

